Лекция 11

10 марта 2026 г. 8:55

Опр. f интегрируема на $[a,b]$ по опр. инт. Римана будем обозначать $\int_a^b f(x)\,dx$

Утв. Пусть f и $g: [a,b] \to \mathbb{R}$ и обе интегрируемы на $[a,b] \Rightarrow \forall \alpha,\beta \in \mathbb{R}$ $\alpha f + \beta g$ интегрируема на $[a,b]$, более того $\int_a^b (\alpha f + \beta g)\,dx = \alpha \int_a^b f\,dx + \beta \int_a^b g(x)\,dx$

Док-во. Для разбиения T и выборки ξ_T

$$G(\alpha f + \beta g, T, \xi_T) = \sum_{k=1}^{n} (\alpha f(\xi_k) + \beta g(\xi_k))(x_k - x_{k-1}) =$$

$$= \alpha \sum_{k=1}^{n} f(\xi_k)(x_k - x_{k-1}) + \beta \sum_{k=1}^{n} g(\xi_k)(x_k - x_{k-1}) =$$

$$= \alpha G(f, T, \xi_T) + \beta G(g, T, \xi_T)$$

$\lambda(T) \to +0$ $\quad$ $\lambda(T) \to +0$ $\quad$ $\lambda(T) \to +0$

$\int_a^b f\,dx$ $\quad$ $\int_a^b g\,dx$ $\quad$ $\Downarrow$

$$\lim_{\lambda(T)\to+0} G(\alpha f + \beta g, T, \xi_T) = \alpha \int_a^b f(x)\,dx + \beta \int_a^b g(x)\,dx$$

Утв. f интегр. на $[a,b] \Rightarrow \forall [\alpha,\beta] \subset [a,b]$ f интегр. на $[\alpha,\beta]$

Док-во: Пусть T разбиение $[\alpha,\beta]$

T_1 разбиение $[a,b]$

T_1 содержит все точки T и $\rho(T_1) = \rho(T)$

$\Delta(f,T) \le \Delta(f,T_1) \xrightarrow[\rho(T)\to+0]{} 0 \Rightarrow f$ интегр. на $[\alpha,\beta]$

1. крит. интегрируемости

Утв. f интегр. на $[a,b]$

$g: [a,b] \to \mathbb{R}$ $\quad \exists \{\xi_k\}_{k=1}^{N} \subset [a,b]$:

$\forall x \in [a,b] \setminus \{\xi_k\} \hookrightarrow g(x) = f(x)$

$\Rightarrow g$ интегрируема на $[a,b]$, более того, $\int_a^b g\,dx = \int_a^b f\,dx$

Док-во: $h(x) = g(x) - f(x)$

$\forall x \in \{\xi_k\}_{k=1}^{N} \hookrightarrow h(x) = 0$

$M = \max_{k\in\{1,\dots,N\}} |h(\xi_k)|$

T-разбиение $[a,b]$ $\quad \Delta(h,T) \le 4MN\,\rho(T) \xrightarrow[\rho(T)\to+0]{} 0$

$\sum \omega_i(h)(x_i - x_{i-1})$

$2M$ $\quad$ $\rho(T)$

$\Rightarrow$ По $\exists$ кр интегрируемости $h(x)$ интегрируема на $[a,b]$

$g = h + f$ интегр. на $[a,b]$

$$\int_a^b g(x)\,dx = \int_a^b f(x)\,dx + \int_a^b h(x)\,dx$$

$\sum M_k(x_k - x_{k-1})$

$$S(h,T) \le 2MN\,\rho(T) \xrightarrow[\rho(T)\to+0]{} 0$$

$$s(h,T) \xrightarrow[\rho\to+0]{} 0 \quad \Rightarrow \int_a^b h(x)\,dx = 0$$

Утв. f интегр на $[a,b] \Rightarrow |f|$ интегр. на $[a,b]$ и $\left|\int_a^b f(x)\,dx\right| \le \int_a^b |f(x)|\,dx$

Док-во. $\|f(x)| - |f(y)\| \le |f(x) - f(y)|$

$\omega_x(|f|) \le \omega_x(f)$

$0 \le \Delta(|f|,T) \le \Delta(f,T) \xrightarrow[\rho(T)\to+0]{} 0$

$\rho(T)\to+0$

0

$\Rightarrow |f|$ интегрируема на $[a,b]$

$G(f,T,\xi_T) \in G(|f|,T,\xi_T)$

$$\left|\sum_{k=1}^{n} f(\xi_k)(x_k - x_{k-1})\right| \le \sum_{k=1}^{n} |f(\xi_k)(x_k - x_{k-1})| = \sum_{k=1}^{n} |f(\xi_k)|(x_k - x_{k-1})$$

$\rho(T)\to+0$ $\quad$ $\int_a^b |f(x)|\,dx$

Утв. f интегр. на $[a,b]$ и на $[b,c]$ $\Downarrow$ f интегр. на $[a,c]$ и $\int_a^c f(x)\,dx = \int_a^b f(x)\,dx + \int_b^c f(x)\,dx$

Док-во. T разбиение $[a,c]$, $b \in [x_i, x_{i+1}]$

$T_1 = \{x_k\}_{k=0}^{i} \cup \{b\}$

$T_2 = \{x_k\}_{k=i+1}^{n} \cup \{b\}$

T_1 разбиение $[a,b]$, T_2 разбиение $[b,c]$

$\rho(T_1) \le \rho(T)$. $\rho(T_2) \le \rho(T_2)$

$$S(f,T) = \sum_{k=1}^{n} M_k(x_k - x_{k-1}) + M_{i+1}(x_{i+1} - x_i),\quad S(f,T_1) = \sum_{k=1}^{i} M_k(x_k - x_{k-1})$$

$$S(f,T_2) = \sum_{k=i+1}^{n} M_k(x_k - x_{k-1}) + \sup_{[b,x_{i+1}]} f(x)(x_{i+1} - b) + \sup_{[b,x_{i+1}]} f(x)(b - x_i)$$

$$|S(f,T) - S(f,T_1) - S(f,T_2)| =$$

$$= \left|M_{i+1}(x_{i+1} - x_i) - \sup_{[x_i,b]} f(x)(b - x_i) - \sup_{[b,x_{i+1}]} f(x)(x_{i+1} - b)\right| =$$

$$= \left|(x_{i+1} - b)\left(M_{i+1} - \sup_{[b,x_{i+1}]} f(x)\right) + (b - x_i)\left(M_{i+1} - \sup_{[x_i,b]} f(x)\right)\right| \le$$

$$\le |(x_{i+1} - b)|\left|M_{i+1} - \sup_{[b,x_{i+1}]} f(x)\right| + |(b - x_i)|\left|M_{i+1} - \sup_{[x_i,b]} f(x)\right| \le$$

$$\le \rho(T)\left(|M_{i+1}| + \left|\sup_{[b,x_{i+1}]} f(x)\right| + \left|\sup_{[x_i,b]} f(x)\right|\right) \le$$

f интегр. на $[a,b] \subset [b,c] \Rightarrow f$ огр. на $[a,b] \cup [b,c]$

$\Rightarrow \exists M: \forall x \in [a,c] \hookrightarrow |f(x)| \in M$

$\rho(T) \xrightarrow[\rho(T)\to+0]{} 0$

$$S(f,T) \xrightarrow[\rho(T)\to+0]{} \int_a^b f(x)\,dx + \int_b^c f(x)\,dx$$

Аналогично для $s(f,T) \xrightarrow[\rho(T)\to+0]{} \int_a^b f(x)\,dx + \int_b^c f(x)\,dx$

Опр. $\int_a^a f(x)\,dx = 0$

если f интегрируема на $[a,b]$, то $\int_a^c f(x)\,dx = -\int_c^a f(x)\,dx$

Утв. f интегр на отрезке, содержащем $a, b, c \Rightarrow$

$$\int_a^c f(x)\,dx = \int_a^b f(x)\,dx + \int_b^c f(x)\,dx$$

c, a, b

$$\int_c^b f(x)\,dx = -\int_b^c f(x)\,dx + \int_a^b f(x)\,dx$$

$$-\int_b^a f(x)\,dx = -\int_b^c f(x)\,dx + \int_a^c f(x)\,dx$$

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$$

Утв. $f: [a,b] \to \mathbb{R}$ $\Rightarrow$ f интегр. на $[a,b]$ $\quad$ f непр. на $[a,b]$

Док-во. T разбиение $[a,b]$, $\omega_k(f) \le \omega_k(2\lambda(T))$

$\sup_{x_i, y_i \in [x_{k-1}, x_k]} |f(x_i) - f(y_i)| \le \sup_{x,y\in[a,b],\ |x-y|\le 2\lambda(T)} |f(x) - f(y)|$

f непр. на $[a,b] \Rightarrow f$ P.H.П. на $[a,b]$

$$\Delta(f,T) = \sum_{k=1}^{n} \omega_k(f)(x_k - x_{k-1}) \le \sum_{k=1}^{n} \omega_k(2\lambda(T))(x_k - x_{k-1}) =$$

$$= \omega_\varepsilon(2\lambda(T))(b - a) \xrightarrow[\lambda(T)\to+0]{} 0$$

Критерий РИМ